

VisiSlide: A Multimodal Gesture and Attention-Driven Presentation Interface

Subject: Human Computer Interaction - Assignment 2

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I. LITERATURE REVIEW SECTION

Paper 1: Eye-gesture control of computer systems via AI (Siva & Selvi, 2024)

- **Problem:** People who are living or surviving with profound motor disabilities are not able to work with the computer with the help of traditional input devices like mouse, keyboard, joystick and stylus.
- **Methodology:** The paper specifically writes about mapping eye movements of the humans to system commands with the help of gaze estimation model based on (DL).
- **Dataset/System:** The study used the custom dataset of eye-tracking images, and in the result the system successfully achieved 99.63% accuracy.
- **Evaluation:** The authors tested the model by checking the behavior of the system by giving the input of different people in the dim or bright lights.
- **New Contribution:** High-accuracy eye-command mapping without the need for specialized, expensive hardware.
- **Limitation:** The model has to perform a lot of work so one of the limitation of the paper under discussion is CPU overhead; another issue could be that the system may generate command of click the button just because the person looked at the button for a few seconds whereas the user had not intent to do this.

Paper 2: Hand Gesture Controlled Presentation System (Rajanala et al., 2026)

- **Problem:** Presenters have to hold the remote control or have to keep their hand onto the mouse or trackpad for controlling the movement of slides. Due to that issue the presenters may lose their contact and the charm of discussion with the audience.
- **Methodology:** It actually uses the libraries of OpenCV for image processing and MediaPipe for the sake of 21-point precise landmark detection.
- **Dataset/System:** Real-time webcam feed analyzed via Python-based ML models for slide navigation.
- **Evaluation:** For testing the accuracy the system the team evaluated the model by controlling the slides through hand gestures from the different distances in different light effects.

- **New Contribution:** The study introduced the “gesture dictionary” translated hand gestures into the actions that can be performed for the movement of slides also it helps the users to draw/write on the presentable screen by tracing the movement of hands in air.
- **Limitation:** The system does not work accurately in the low light environment and it does not have any verification layer to verify the user really want to perform any action or that was just a body language.

Paper 3: Mapping Eye-Tracking Research in HCI (MDPI, 2026)

- **Problem:** The delivery of content and navigation of interface at the same time may create a problem of loss of focus and cognitive load for the presenters.
- **Methodology:** An analysis of trends in Human-Computer Interaction (HCI), specifically at the confluence of deep learning, extended reality, and eye-tracking metrics.
- **Dataset/System:** Four main models of interactions identified through a comprehensive meta-analysis of HCI literature.
- **Evaluation:** Conducted evaluation to determine whether the proposed approach could help in alleviating mental fatigue and obtaining effective systems based on the concept of gaze.
- **New Contribution:** Identifies eye-tracking as a “core enabling technology” for reducing chronic mental fatigue in users.
- **Limitation:** The study is just based onto the theory and it did not provide the complete implementation logic for the real 2D presentation platform.

Paper 4: Dynamic Hand Gesture Recognition Using Transformer Networks (Malleswara Rao, 2025)

- **Problem:** The traditional hand gesture system suffers with the problem of “Midas Touch” In which unintentional hand movement is interpreted as the command.
- **Methodology:** The method makes use of transformer-based models to assess time series data related to motions.
- **Dataset/System:** MediaPipe-extracted landmarks processed through a sequence-to-sequence model (KArSL datasets).

- **Evaluation:** Evaluated by dynamic vs static detection accuracy across multiple classes.
- **New Contribution:** Applies “dynamic swipes” for differentiating between deliberate gestures and random motions that occur while speaking.
- **Limitation:** Longer response times and greater complexity in model design compared to traditional ANN systems.

Paper 5: A Visual Attention-based Method for the Midas Touch Problem (ResearchGate)

- **Problem:** The main challenge is differentiating between the body language of the speaker and the formal control command given by the speaker.
- **Methodology:** Constructs a “parallel perception model” combining top-down and bottom-up attention to verify intent.
- **Dataset/System:** Visual attention tracking integrated with a gesture toolkit to facilitate dynamic spotting.
- **Evaluation:** Authors tested the system in Virtual Reality by counting how many mistakes it made and asking users how much effort it took to use, then compared those results to older methods.
- **New Contribution:** Added the gaze validation layer makes sure that commands are only executed when the users have intention to run them.
- **Limitation:** The discussed framework is only focused to just VR/AR environment. But the more modifications in this way may generate an operating software for controlling 2D presentation software on laptop.

Paper 6: Real-time Hand Gesture Recognition using MediaPipe and LSTM

- **Problem:** Traditional ML classifiers usually treat gestures like photo frames that actually lose the movement (flow) of gestures which is required for complex commands.
- **Methodology:** Pairs MediaPipe (for hand tracking) with LSTM neural networks (for “remembering” movement patterns).
- **Dataset/System:** KeyPoint analysis of spatial-temporal movements (DHG-14/28 and SHREC datasets).
- **Evaluation:** Evaluated the system by using the F1 score making sure that the systems works well with both sign language and computer interaction.
- **New Contribution:** It does not only look for the shape of the hand it actually stores the movement of hand in the long term memory of LSTM.
- **Limitation:** Scalability becomes an issue since adding any new gesture means acquiring a lot of data.

Paper 7: Multimodal Gesture and Speech-Based Interaction Interfaces

- **Problem:** Single-modality systems often suffers in the dynamic environment (real world) due to visual clutter or the movement of background.
- **Methodology:** Fuses real-time computer vision (CNN-LSTM) with NLP models for cross-modal verification.

- **Dataset/System:** ROS-based middleware for robot control, allowing for “confirmative” speech gating.
- **Evaluation:** Achieved an extremely high accuracy rate of 92.7% in the busy environment with multiple people in surroundings.
- **New Contribution:** Succeeded to develop a hybrid co-operative environment where when the camera misses the gesture the microphone will capture the voice (vice versa).
- **Limitation:** The system requires two active environments at the same time for smooth processing also require high processing for the detection of gesture in the populated visual environment.

Paper 8: AI Virtual Mouse System: Revolutionizing HCI (SciTePress, 2025)

- **Problem:** Normal mechanical mice may be inaccessible to those suffering from neuromuscular disabilities, as well as being impractical in mobile situations.
- **Methodology:** Uses color recognition and fingertip detection via standard webcams to emulate mouse functions.
- **Dataset/System:** OpenCV-based real-time filtering and conversion of coordinates to screen pixels.
- **Evaluation:** The efficiency of the system was measured using cursor accuracy, system response time, and user involvement.
- **New Contribution:** Achieves a 100% software-based interaction model, eliminating all secondary hardware dependencies.
- **Limitation:** The cursor control might be shaky or abnormal due to different skin color shades and lighting effects.

Paper 9: Human centric heuristics in gesture recognition (IJERT/IJIRT)

- **Problem:** “Intent Gap” where AI systems perform those actions that user did not intend consciously.
- **Methodology:** To tackle above problem authors propose non-verbal communication heuristics specifically using hand tracking algorithms.
- **Dataset/System:** Scientists utilized HaGRID dataset used in training across various scenes and subjects effectively.
- **Evaluation:** They evaluated how well users engaged with the system in educational settings as compared to traditional clickers.
- **New Contribution:** Focuses on bridging gap between human intuition and machine actions by adding the heuristic layers.
- **Limitation:** Absence of strict mathematical “gating” layer for verification of the focus.

Paper 10: Dynamic hand Gesture recognition using MediaPipe and Transformer (2025)

- **Problem:** Problem with classifying gestures that look similar like a palm and waving hand in pictures.
- **Methodology:** Replaces traditional image processing with MediaPipe to find the key points fed into a 3D Transformer model.

- **Dataset/System:** Created a set of 12 commands for emergency responders to use as signals.
- **Evaluation:** It reduces the noise in background and mistakes in classifying gestures.
- **New Contribution:** Approach uses an encoding method that helps process gestures in order, one after the other.
- **Limitation:** Training the model requires computers with high-end GPUs to handle the complex calculations.

Paper 11: High-Accuracy Gaze Estimation for Interpolation-Based Eye-Tracking

- **Problem:** Eye tracking systems have trouble working accurately because of low quality calibration and small mistakes.
- **Methodology:** Deterministic mathematical checks of pupil to camera vectors and Euclidean distance.
- **Dataset/System:** Datasets that focus on where people are looking.
- **Evaluation:** Accuracy measured in visual angle degrees to ensure the gaze error is minimized.
- **New Contribution:** Provides a method for plug and play eye tracking without extensive user calibration.
- **Limitation:** If the user face is not straight on our system then it does not work accurately.

Paper 12: Zero Calibration Landmark Detection in Low-Light

- **Problem:** Most vision systems fail in environments that are dim like lecture rooms.
- **Methodology:** Use low-light enhancement algorithms to utilize high fidelity landmark detection.
- **Dataset/System:** Tested on the NVGesture dataset which includes infrared and depth modalities.
- **Evaluation:** Robustness testing under varied illumination levels (0-100 lux).
- **New Contribution:** Reliable performance in “low-fidelity environments” using normalized keypoint data.
- **Limitation:** Higher latency due to the pre-processing “enhancement” step.

Paper 13: Temporal Verification of Gesture Trajectories (Li & Hsieh, 2025)

- **Problem:** Accidental flicks or hand movements during speech cause slide skips.
- **Methodology:** Math to look at how fast people’s hands were moving over a period of time like 30 frames.
- **Dataset/System:** Sequence-based analysis system implemented via MediaPipe landmarks.
- **Evaluation:** Comparison of “intentional flick” vs “natural talk” gesture speeds.
- **New Contribution:** Fast and deliberate movements should cause the slides to change.
- **Limitation:** If the threshold for speed is set high it might miss slow gestures.

Paper 14: GFIE: A Dataset and Baseline for Gaze-Following (CVPR, 2023)

- **Problem:** Lack of big, unbiased datasets to train AI to understand what people are looking at.

- **Methodology:** Developed a gaze data collection system that uses laser spots to get people to look at things.
- **Dataset/System:** GFIE (Gaze-Following in Indoor Environments) dataset with 2D/3D gaze targets.
- **Evaluation:** Compared against the CAD-120 dataset for intention prediction accuracy.
- **New Contribution:** Automatically labels data so that it is not biased by people’s opinions.
- **Limitation:** Specifically optimized for indoor environments; may struggle with outdoor glare.

Paper 15: AI Testing and Evaluation for Accessibility (PMC, 2026)

- **Problem:** AI systems often act as black boxes, creating barriers for users with different physical needs.
- **Methodology:** A comprehensive framework for AI testing, evaluation, verification, and validation (TEVV).
- **Dataset/System:** Different test datasets representing a wide range of disabilities.
- **Evaluation:** Field testing and usability metrics focused on inclusivity and bias detection.
- **New Contribution:** A way to standardize the verification layer for AI tools focused on accessibility.
- **Limitation:** Framework is highly theoretical and requires custom work to implement in apps.

II. COMPARATIVE ANALYSIS TABLE

TABLE I
COMPARISON OF METHODOLOGIES

Paper Modality	Key Contribution	Evaluation Method	Limitation
Eye-Gesture (AI)	99.63% Accuracy	Accuracy vs. Light	High CPU Load
MediaPipe/OpenCV	21-point Landmarks	Distance/Light tests	Range < 300cm
Transformer Net	Temporal sequences	Task load study	High Latency
Hybrid CNN-LSTM	Multi-modal Fusion	Noise-robustness	System Complexity
Visual Attention	Gaze-based Verification	Error count study	Focused on VR
VisiSlide (Proposed)	Gaze-Driven Logic Gate	Intent-Aware Testing	Requires Frontal View

III. REFERENCES

A. Core Research Papers (Presentation & Gestures)

- 1) Eye-gesture control of computer systems via AI (2024). *Focus: High-accuracy eye-command mapping.*
- 2) Hand Gesture Controlled Presentation System (2026). *Focus: OpenCV and MediaPipe implementation for slide navigation.*
- 3) Mapping Eye-Tracking Research in HCI (2026). *Focus: Meta-analysis on reducing cognitive load.*
- 4) Dynamic Hand Gesture Recognition Using Transformer Networks (2025). *Focus: Using temporal sequences to solve the “Midas Touch.”*

- 5) A Visual Attention-based Method for the Midas Touch Problem. *Focus: Parallel perception models for intent verification.*

B. Technical Implementation & Datasets

- 6) Real-time Hand Gesture Recognition using MediaPipe and LSTM. *Focus: Capturing spatial-temporal trajectories of hand movements.*
- 7) Multimodal Gesture and Speech-Based Interaction Interfaces (2025). *Focus: Hybrid deep learning (CNN-LSTM) for sensor fusion.*
- 8) AI Virtual Mouse System: Revolutionizing HCI (2025). *Focus: Fingertip detection for hardware-free control.*
- 9) Human-Centric Heuristics in Gesture Recognition. *Focus: Reducing the “Intent-Gap” in non-verbal communication.*
- 10) GFIE: A Dataset and Baseline for Gaze-Following (CVPR 2023). *Focus: Large-scale 2D/3D gaze targets in indoor environments.*

C. Mathematical Foundations (Logic Gating)

- 11) High-Accuracy Gaze Estimation for Eye-Tracking. *Focus: Point-of-Regard (PoR) precision.*
- 12) Zero Calibration Landmark Detection in Low-Light. *Focus: Robustness in dim-lit presentation environments.*
- 13) Temporal Verification of Gesture Trajectories (2025). *Focus: 30-frame temporal window analysis for “flicks.”*
- 14) GazeD: Context-Aware Diffusion for Accurate 3D Gaze Estimation (2026). *Focus: Predicting human pose and intention from a single image.*
- 15) AI Testing and Evaluation for Accessibility (2026). *Focus: Verification and Validation (TEVV) framework.*